

KEARNEY, NE, USA

WMO: 725526

Lat: 40.733N

Lon: 99.000W

Elev: 2130

StdP: 13.60

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14905

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	2.8	-12.8	3.0	1.9	-6.5	4.2	4.5	33.0	29.7	29.0	28.5	10.0	320	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	94.1	73.1	91.1	72.7	89.4	72.1	77.0	89.0	75.4	87.0	73.7	85.2	14.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.1	133.1	84.7	72.0	128.2	83.7	70.1	120.0	81.6	42.3	89.1	40.3	87.2	38.7	85.0	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.6	26.8	24.2	-9.4	99.4	4.9	2.3	-13.0	101.0	-15.9	102.4	-18.6	103.7	-22.2	105.4
			-10.5	80.4	4.9	2.0	-14.0	81.8	-16.9	83.0	-19.6	84.1	-23.2	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	26.3	29.3	40.0	49.9	60.6	71.2	75.8	73.2	65.8	52.1	38.8	28.2
	DBStd	19.92	11.59	12.42	11.67	9.80	8.52	6.82	5.51	5.36	8.14	9.33	10.27	10.62
	HDD50	2913	735	584	346	119	12	0	0	0	3	85	354	675
	HDD65	6078	1200	1000	777	460	190	22	3	5	87	409	785	1140
	CDD50	3295	0	3	35	116	340	636	800	718	478	150	19	0
	CDD65	985	0	0	1	7	52	208	338	258	112	9	0	0
	CDH74	10087	0	1	30	151	653	2167	3417	2299	1202	161	6	0
	CDH80	4109	0	0	4	39	218	877	1554	926	449	42	0	0
(14) Wind	WSAvg	11.3	10.8	11.7	12.7	13.9	12.7	11.7	9.3	8.9	10.7	11.4	11.3	10.6
(15) Precipitation	PrecAvg	26.1	0.5	0.5	1.5	2.2	4.4	4.2	3.4	3.6	1.8	2.4	0.9	0.5
	PrecMax	38.5	1.4	1.5	3.2	4.8	8.2	7.6	12.3	7.4	4.8	9.3	2.4	1.7
	PrecMin	15.4	0.1	0.0	0.0	0.5	1.2	1.5	0.1	1.0	0.3	0.3	0.0	0.1
	PrecStd	5.4	0.4	0.4	1.0	1.1	2.0	1.8	2.7	1.6	1.3	1.9	0.7	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	67.8	78.8	84.4	91.0	97.0	99.3	97.1	93.4	85.6	73.2	60.6
		MCWB	45.1	49.3	55.6	59.3	66.1	71.7	74.3	73.2	68.7	62.8	53.5	45.0
	2%	DB	53.5	59.2	72.0	78.8	86.1	91.7	94.8	91.5	89.8	78.8	66.1	53.8
		MCWB	41.6	44.9	53.4	58.1	66.3	71.9	74.8	73.9	69.3	59.7	50.4	43.1
	5%	DB	46.4	53.1	64.4	72.6	81.4	89.6	91.2	89.6	85.2	72.9	61.0	47.7
		MCWB	37.7	42.7	50.7	56.1	63.5	71.1	74.5	73.8	67.9	57.2	47.6	39.4
	10%	DB	42.6	46.2	59.0	66.9	76.8	84.4	89.0	84.6	81.3	68.3	54.8	43.3
		MCWB	35.5	38.6	47.5	53.4	61.8	69.1	73.5	71.8	65.7	55.5	45.1	36.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	50.8	60.4	63.2	71.1	77.4	79.7	78.7	74.3	66.7	56.6	49.2
		MCDB	59.5	64.7	70.7	75.6	83.2	89.1	92.4	89.5	86.4	78.5	66.9	59.5
	2%	WB	41.5	46.1	56.0	60.2	68.3	74.7	77.4	76.4	71.6	63.0	52.3	42.8
		MCDB	51.1	57.3	66.9	73.7	81.3	87.0	90.0	87.4	84.3	72.9	63.3	52.3
	5%	WB	38.5	42.6	51.9	57.3	66.0	72.4	76.0	74.8	69.6	59.6	49.0	39.8
		MCDB	46.0	52.4	63.9	71.2	78.5	85.5	88.1	85.6	81.6	69.2	60.1	47.5
	10%	WB	35.5	38.8	48.0	54.6	63.3	70.4	74.4	72.9	67.6	56.7	45.4	36.5
		MCDB	41.6	46.1	59.5	65.7	73.8	83.3	86.4	83.3	78.3	66.9	54.6	43.1
(35) Mean Daily Temperature Range	5% DB	MDBR	20.1	21.1	24.1	23.8	21.9	22.1	22.0	21.5	24.4	24.2	22.9	19.8
		MCDBR	27.6	30.6	34.4	32.9	28.7	27.1	26.4	26.7	29.2	31.7	31.9	27.2
		MCWBR	18.5	19.3	19.7	17.6	13.7	12.2	12.6	12.5	13.1	16.5	19.1	18.3
	5% WB	MCDBR	26.4	28.7	31.2	29.4	25.5	24.5	23.7	23.3	25.3	26.2	30.2	26.2
		MCWBR	18.3	18.8	18.5	16.8	13.4	12.7	12.7	12.5	12.6	15.6	18.8	18.4
(40) Clear-Sky Solar Irradiance	taub		0.272	0.274	0.300	0.340	0.366	0.381	0.396	0.392	0.357	0.313	0.275	0.266
	taud		2.494	2.509	2.466	2.370	2.374	2.374	2.363	2.361	2.413	2.519	2.542	2.510
	Ebn at Noon		286	303	304	296	289	283	278	276	279	283	284	280
	Edh at Noon		23	27	31	37	38	38	38	37	33	26	22	21
(44) All-Sky Solar Radiation	RadAvg		711	974	1325	1664	1924	2210	2168	1886	1530	1096	785	608
	RadStd		64	70	113	92	122	135	94	103	93	109	54	57

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	+0.63	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page